The electromagnetic spectrum

antocciphysics | TEACHER KEY | CPA

Learning target: Relate electromagnetic wavelength, frequency and propagation.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 9; 3 course days later.*
50 m; one-way distance is half the round trip.

02. Explain a worked example (worked)
Shorter wavelength has higher frequency; different EM types do not require different vacuum speeds.

03. Look and explain (model)
Radio, visible, X-ray; X-ray highest frequency.

04. Build the solution (practice)
4 m; halves.

05. Use the idea (reason)
EM waves do not require material medium; ordinary sound is pressure motion of matter.

06. Choose a method (mixed) *Review: Unit 5, Day 12; 12 course days later.*
+40 J; internal energy increases.

07. Exit check (exit)
No; frequency and wavelength change.

08. STRETCH: Apply it another way (stretch)
Examples 10^6 Hz and 300 m; 10^9 Hz and 0.3 m.

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01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 9; 3 course days later.*
$\lambda = 1.2$ m; $f \approx 283.3$ Hz.

02. Explain a worked example (worked)
Shorter wavelength has higher frequency; different EM types do not require different vacuum speeds.

03. Represent and interpret (model)
Radio, visible, X-ray; X-ray highest frequency.

04. Solve and justify (practice)
5×10^{14} Hz; 3 m.

05. Analyze the reasoning (reason)
EM waves do not require material medium; ordinary sound is pressure motion of matter.

06. Choose a method (mixed) *Review: Unit 5, Day 12; 12 course days later.*
Carnot limit $\eta \approx 0.3333$; $W_{max} \approx 266.7$ J. Irreversibility and losses reduce actual useful work.

07. Exit check (exit)
0.125 m; microwave.

08. STRETCH: Apply it another way (stretch)
5×10^{14} Hz; 300 nm gives 10^{15} Hz, twice the frequency.